Algorithm 3 Dijkstra's algorithm on the grid

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1: function DIJKSTRA(s_start, s_goal)
2:     g(s_start) ← 0;  OPEN ← {s_start};  CLOSED ← ∅
3:     while OPEN ≠ ∅ do
4:         s ← arg min_{s∈OPEN} g(s)
5:         if s = s_goal then
6:             return RECONSTRUCTPATH(parent, s_goal)
7:         end if
8:         OPEN ← OPEN \ {s};  CLOSED ← CLOSED ∪ {s}
9:         for all s' ∈ NEIGHBORS(s) do
10:            if s' ∈ CLOSED ∨ ¬COLLISIONFREE(s, s') then
11:                continue
12:            end if
13:            g' ← g(s) + COST(s, s')
14:            if s' ∉ OPEN ∨ g' < g(s') then
15:                g(s') ← g';  parent(s') ← s
16:                OPEN ← OPEN ∪ {s'}
17:            end if
18:        end for
19:    end while
20:    return failure
21: end function
```